Clearance & Elimination Practice 4

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

1a 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. a. What is the probable mechanism of renal excretion of this drug?

A. Hepatic metabolism

B. Filtration with active secretion

C. Filtration with tubular reabsorption

D. Filtration only

1b 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. b. What is the usual t½ for this drug?

1c 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. c. What would be the new t½ for this drug in an adult with partial renal failure whose clearance of the antibiotic was 125 mL/min?

Problem 2 Estimate the CrCl of a 42-year-old female who weighs 82kg, S_{Cr} =0.9 mg/dL, and 170 cm tall.